

Counting (Part II)

More Complicated Counting Problems

- ① A peace organization has 8 doctors, 15 nurses, 5 scientists, and 10 volunteers (with no special skills). The organization will choose a team of 6 doctors, 9 nurses, 3 scientists, and 5 volunteers to help some sick people in an impoverished country. How many outcome variations are possible for the team that the organization will send?

Choice #1 Choice #2 Choice #3 Choice #4

$$8C_6 \times 15C_9 \times 5C_3 \times 10C_5$$

$$28 \times 5,005 \times 10 \times 252$$

353,152,800 outcome variations

- ② There are 28 band members at Rockridge High School, 33 at Sierra High, and 30 at El Dorado High. The band director from Rockridge will choose 7 students to play in a district "honor" band, the Sierra band director will choose 10 students to play in this same honor band, and the El Dorado director will choose 12 students. How many outcome variations are possible for the honor band (i.e. the 29 students chosen)?

Choice #1 Choice #2 Choice #3

$$28C_7 \times 33C_{10} \times 30C_{12}$$

$$1184040 \times 92,561,040 \times 86,493,225$$

$9.479309221 \times 10^{21}$ outcome variations

- ③ Corey has 36 baseball cards, 24 football cards, 18 hockey cards, 30 Pokémon cards, and 20 basketball cards. **Each card is unique.** He wants to show some of his cards to his fellow students in school, but he does not want to take all of them. He decides to take 4 baseball cards, 2 football cards, 5 hockey cards, 2 Pokémon cards, and 5 basketball cards. How many ways (i.e. outcome variations) can he do this?

Choice #1 Choice #2 Choice #3 Choice #4 Choice #5

$$36C_4 \times 24C_2 \times 18C_5 \times 30C_2 \times 20C_5$$

$$58,905 \times 276 \times 8,568 \times 435 \times 15,504$$

$9.394500998 \times 10^{17}$ outcome variations

- ④ It is Tyler's birthday. His mom said that he can pick 3 of his 5 friends to play video games with for the whole day. His mom also said that he can choose 4 new video games out of the 10 that he likes at the game store, 2 places to order food out of his 6 favorite restaurants, 3 Muscle Man action figures out of the 10 available, and 5 baseball cards out of the 9 that he likes. How many outcome variations are possible?

Choice #1 Choice #2 Choice #3 Choice #4 Choice #5

$$5C_3 \times 10C_4 \times 6C_2 \times 10C_3 \times 9C_5$$

$$10 \times 210 \times 15 \times 120 \times 126$$

476,280,000 outcome variations

⑤ Lily has 12 distinct pairs of pants, 21 distinct shirts, 9 distinct pairs of shoes, and 13 distinct hats. She is packing for her vacation, but she will only take 5 pairs of pants, 7 shirts, 4 pairs of shoes, and 3 hats. How many outcome variations are there?

Choice #1 Choice #2 Choice #3 Choice #4

$${}_{12}C_5 \quad X \quad {}_{21}C_7 \quad X \quad {}_9C_4 \quad X \quad {}_{13}C_3$$

$$792 \quad X \quad 116,280 \quad X \quad 126 \quad X \quad 286$$

$$\boxed{3,318,690,735 \times 10^{12} \text{ outcome variations}}$$

⑥ Jesse is preparing to show his coin collection to his class. He does not want to bring all of his coins, so he will choose some out of his collection. He has 7 distinct dimes, 4 distinct pennies, and 5 distinct quarters. If he decides to bring 2 pennies, 4 dimes, and 3 quarters, how many possible outcome variations are there for his coin presentation? Although some coins have the same denomination (e.g. 4 coins are all pennies), every coin is unique in some way (e.g. One quarter might be made of silver while another quarter could be made of a copper alloy).

Choice #1 Choice #2 Choice #3

$${}_7C_4 \quad X \quad {}_4C_2 \quad X \quad {}_5C_3$$

$$35 \quad X \quad 6 \quad X \quad 10$$

$$\boxed{2,100 \text{ outcome variations}}$$

Sets

Set: A collection of things.

Examples: $\{ \text{People with brown hair} \}$ or $\{ \text{odd #'s} \}$

Sets are often notated with letters such as "A" and "B".

Examples: $A = \{ \text{Birds} \}$, $B = \{ 3, 6, 11, 15 \}$, $C = \{ \text{Athletes} \}$

The Intersection of Sets: If "A" and "B" are sets, then $A \cap B$ is the set containing all things in both "A" and "B".

- ⑦ Let $A = \{ 1, 2, 3, 4, 5 \}$ and $B = \{ 3, 4, 7 \}$.
Find $A \cap B = \boxed{\{ 3, 4 \}}$
- ⑧ Let $A = \{ -6, 7, 18, 5 \}$ and $B = \{ 7, -5, 3, -6, 5 \}$.
Find $A \cap B = \boxed{\{ -6, 5, 7 \}}$
- ⑨ Let $A = \{ -10, 8, 17, 20 \}$ and $B = \{ 100, 3, 22 \}$.
Find $A \cap B = \emptyset$
- ⑩ Let $A = \{ 10, 11, 12, 13, 14 \}$ and $B = \{ 11, 12, 13, 14, 15, 16 \}$.
Find $A \cap B = \boxed{\{ 11, 12, 13, 14 \}}$
- ⑪ Let $A = \{ 100, 99, 201 \}$ and $B = \{ 98, 99, 200 \}$.
Find $A \cap B = \boxed{\{ 99 \}}$
- ⑫ Let $A = \{ 80, 81, 82 \}$ and $B = \{ 79, 80, 81 \}$

Find $A \cap B = \boxed{\{80, 81\}}$

The Union of Sets: If "A" and "B" are sets, then $A \cup B$ is the set containing all of the things in "A" and all of the things in "B".

• (13) Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$

Find $A \cup B = \boxed{\{1, 2, 3, 4, 5, 6, 7\}}$

(14) Let $A = \{-6, 7, 18, 5\}$ and $B = \{7, -5, 3, -6, 5\}$

Find $A \cup B = \boxed{\{-6, -5, 3, 5, 7, 18\}}$

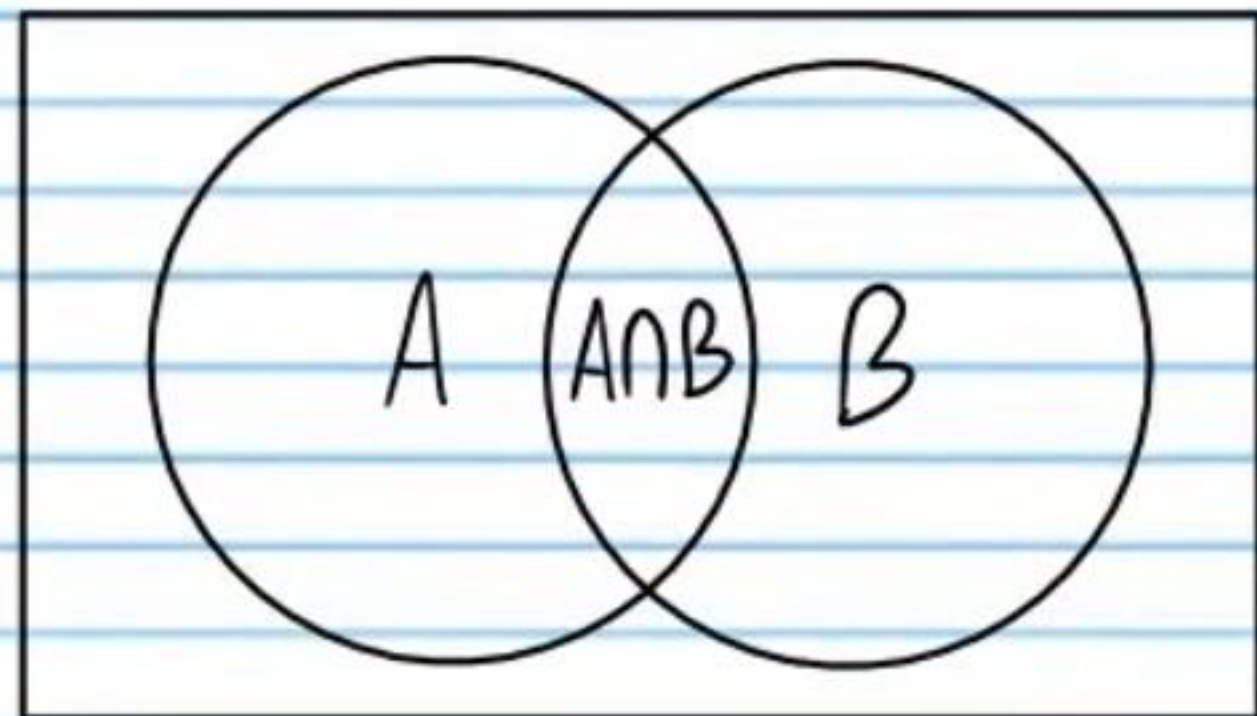
• (15) Let $A = \{-10, 8, 17, 20\}$ and $B = \{100, 3, 22\}$

Find $A \cup B = \boxed{\{-10, 3, 8, 17, 20, 22, 100\}}$

(16) Let $A = \{100, 99, 201\}$ and $B = \{98, 99, 200\}$

Find $A \cup B = \boxed{\{98, 99, 100, 200, 201\}}$

Venn Diagram: A visual representation of groups (i.e. sets) of things in the form of circles and/or rectangles.



Use Venn diagrams to solve the following problems.

- (17) There are 32 students in a class. 20 of them like math, 14 of them like art, and 8 of them like both

i) How many students like math only?

$$20 - 8 = \boxed{12}$$

ii) How many students like art only?

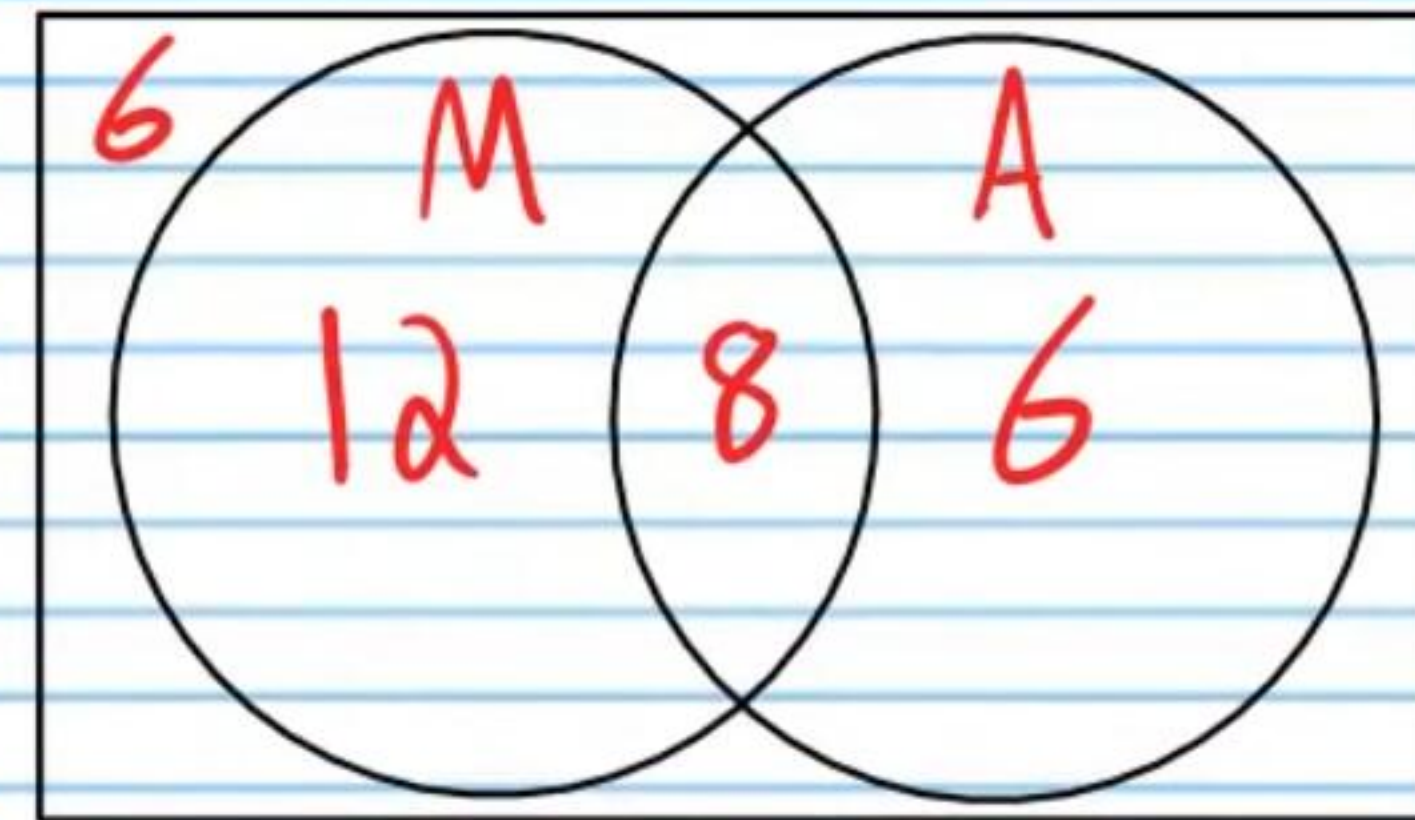
$$14 - 8 = \boxed{6}$$

iii) How many students like **either** math **or** art?

$$12 + 8 + 6 = \boxed{26}$$

iv) How many students like neither math nor art?

$$32 - 26 = \boxed{6}$$



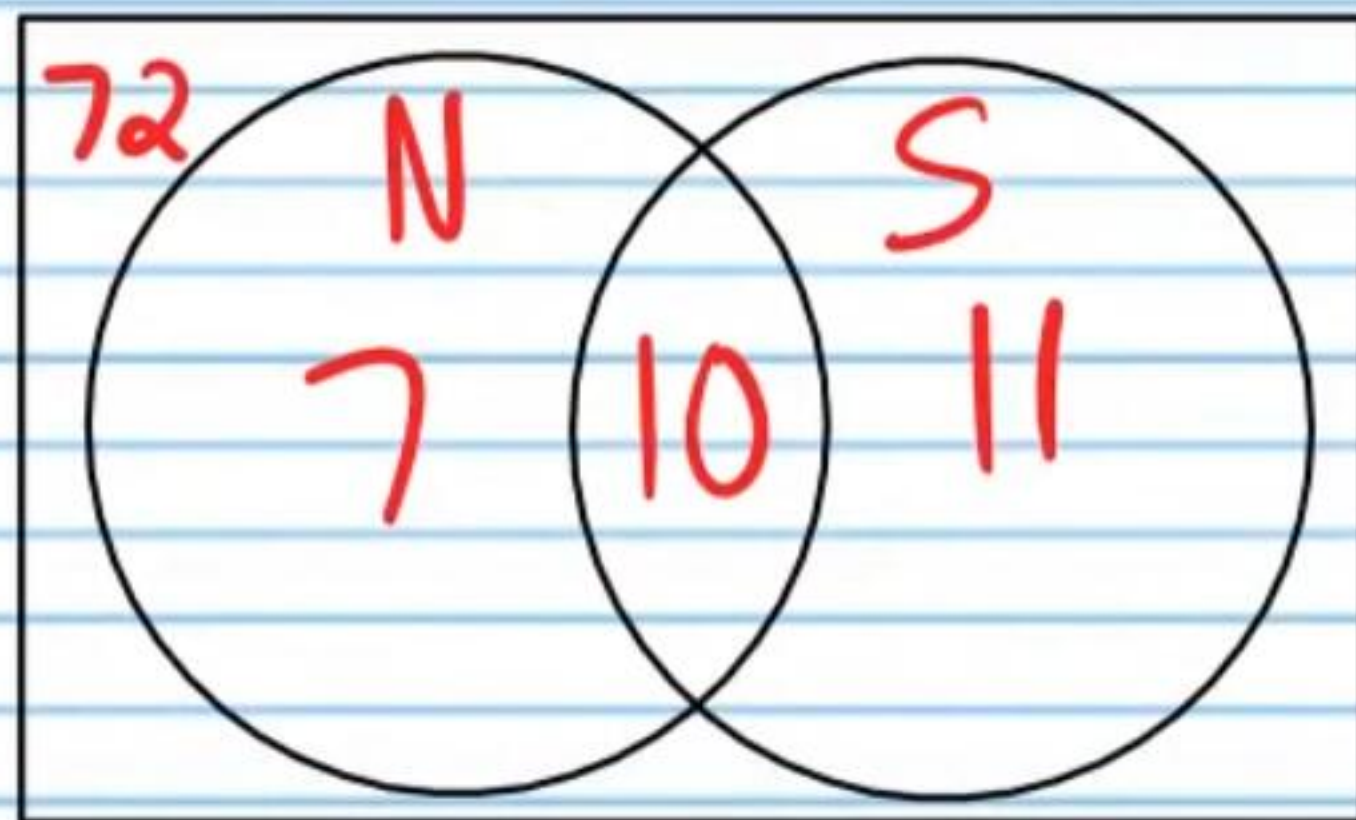
⑮ A tech factory makes 100 calculators one day. 17 of the calculators are nonfunctional, 21 of them have scratches, and 10 of them are nonfunctional with scratches.

i) How many calculators are functional but have scratches?
 $21 - 10 = 11$

ii) How many calculators are nonfunctional but have no scratches?
 $17 - 10 = 7$

iii) How many calculators have **either** scratches **or** are nonfunctional?
 $7 + 10 + 11 = 28$

iv) How many calculators have no scratches and are operational?
 $100 - 28 = 72$



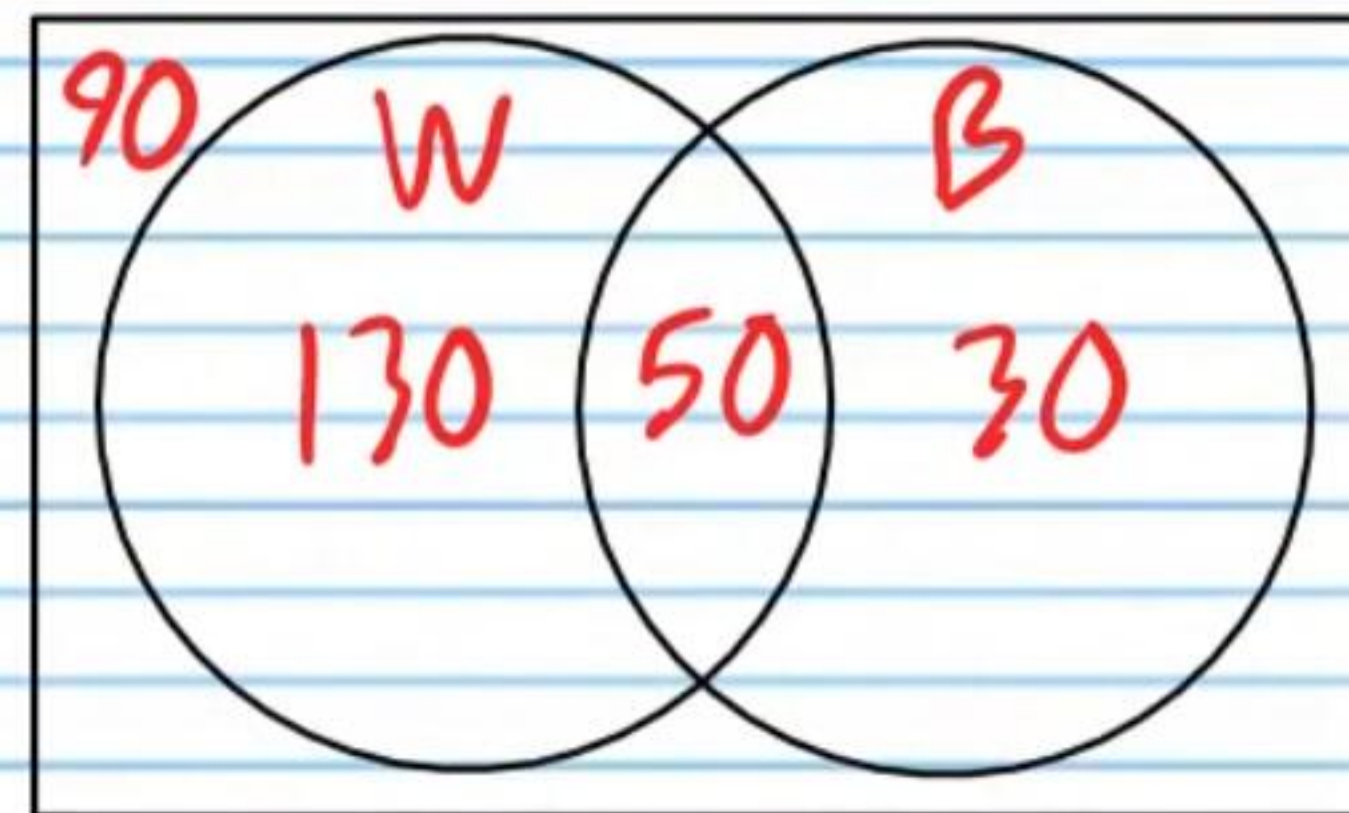
⑯ An aviary has 300 parakeets. 180 have white feathers, 80 have blue feathers, and 50 have white feathers and blue feathers.

i) How many parakeets have white feathers but no blue feathers?
 $180 - 50 = 130$

ii) How many parakeets have blue feathers but no white feathers?
 $80 - 50 = 30$

iii) How many parakeets have **either** white **or** blue feathers?
 $130 + 50 + 30 = 210$

iv) How many parakeets have no white or blue feathers?
 $300 - 210 = 90$



- (20) In one week Ford Motor Company produces 207 trucks. 109 of them have cruise control, 28 of them have sunroofs, and 20 of them have both cruise control and sunroofs.

i) How many trucks have sunroofs but not cruise control?

$$28 - 20 = 8$$

ii) How many trucks have cruise control but not sunroofs?

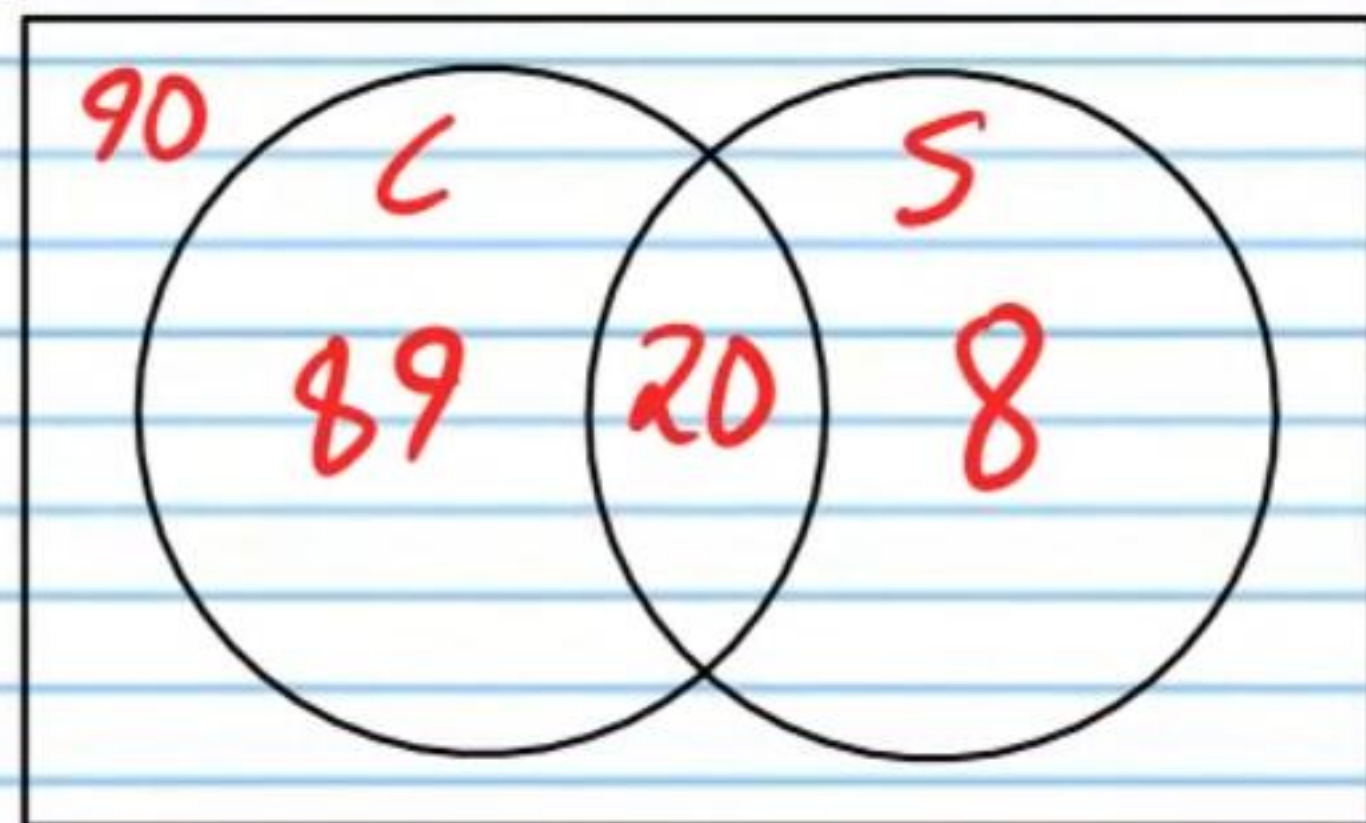
$$109 - 20 = \boxed{89}$$

iii) How many trucks have **either** sunroofs **or** cruise control?

$$89 + 20 + 8 = \boxed{117}$$

iv) How many trucks have neither sunroofs nor cruise control?

$$207 - 117 = \boxed{90}$$



- (21) A family of 36 went on vacation in the mountains. 19 skied and 15 fished. 10 did both.

i) How many family members skied only?

$$19 - 10 = \boxed{9}$$

ii) How many family members fished only?

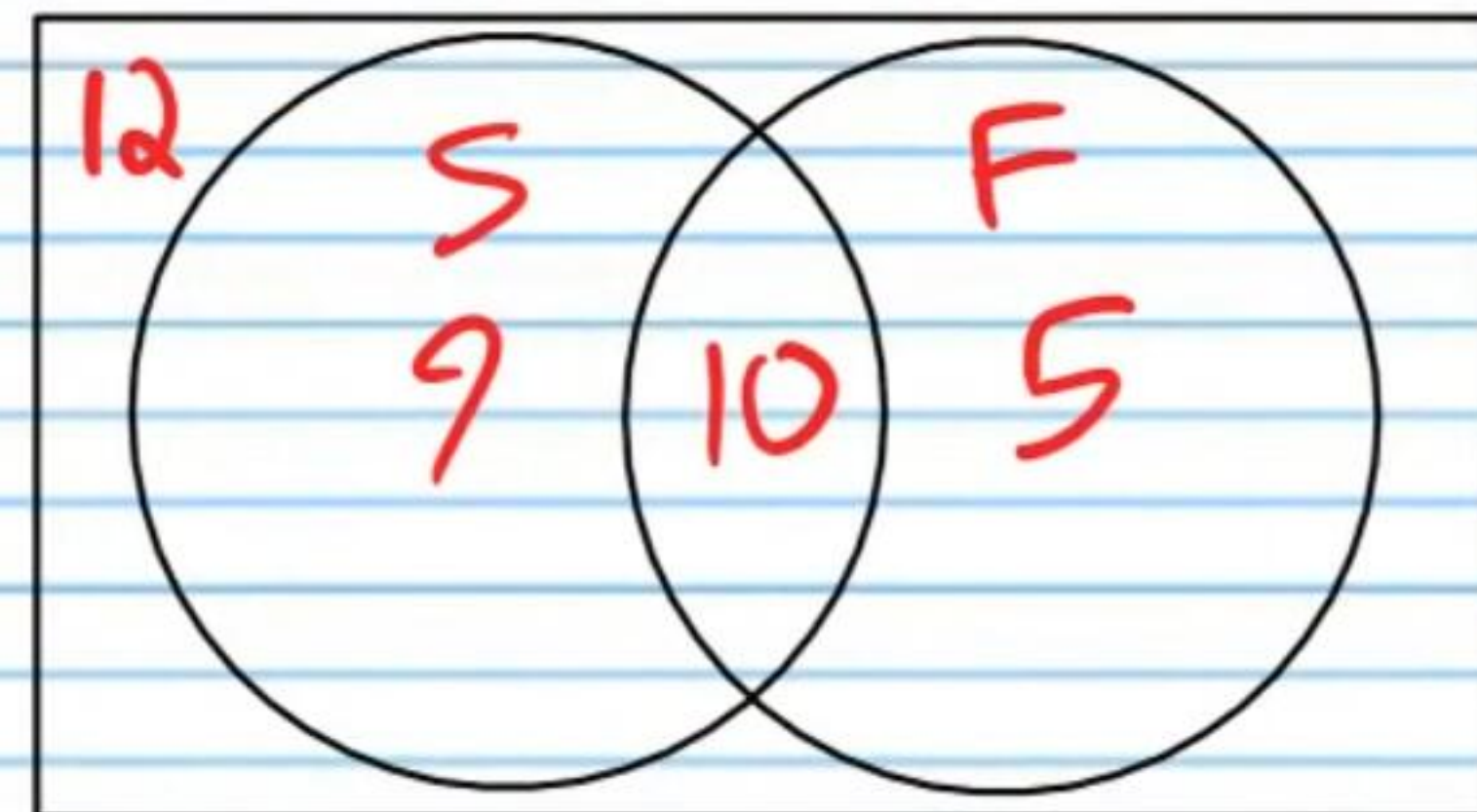
$$15 - 10 = \boxed{5}$$

iii) How many family members **either** fished **or** skied?

$$9 + 10 + 5 = \boxed{24}$$

iv) How many family members neither fished nor skied?

$$36 - 24 = \boxed{12}$$



(2d) There are 100 students in a school. 78 of them like peanut butter. 24 of them like jelly. 14 of them like both.

i) How many students like peanut butter but not jelly?

$$78 - 14 = \boxed{64}$$

ii) How many students like jelly but not peanut butter?

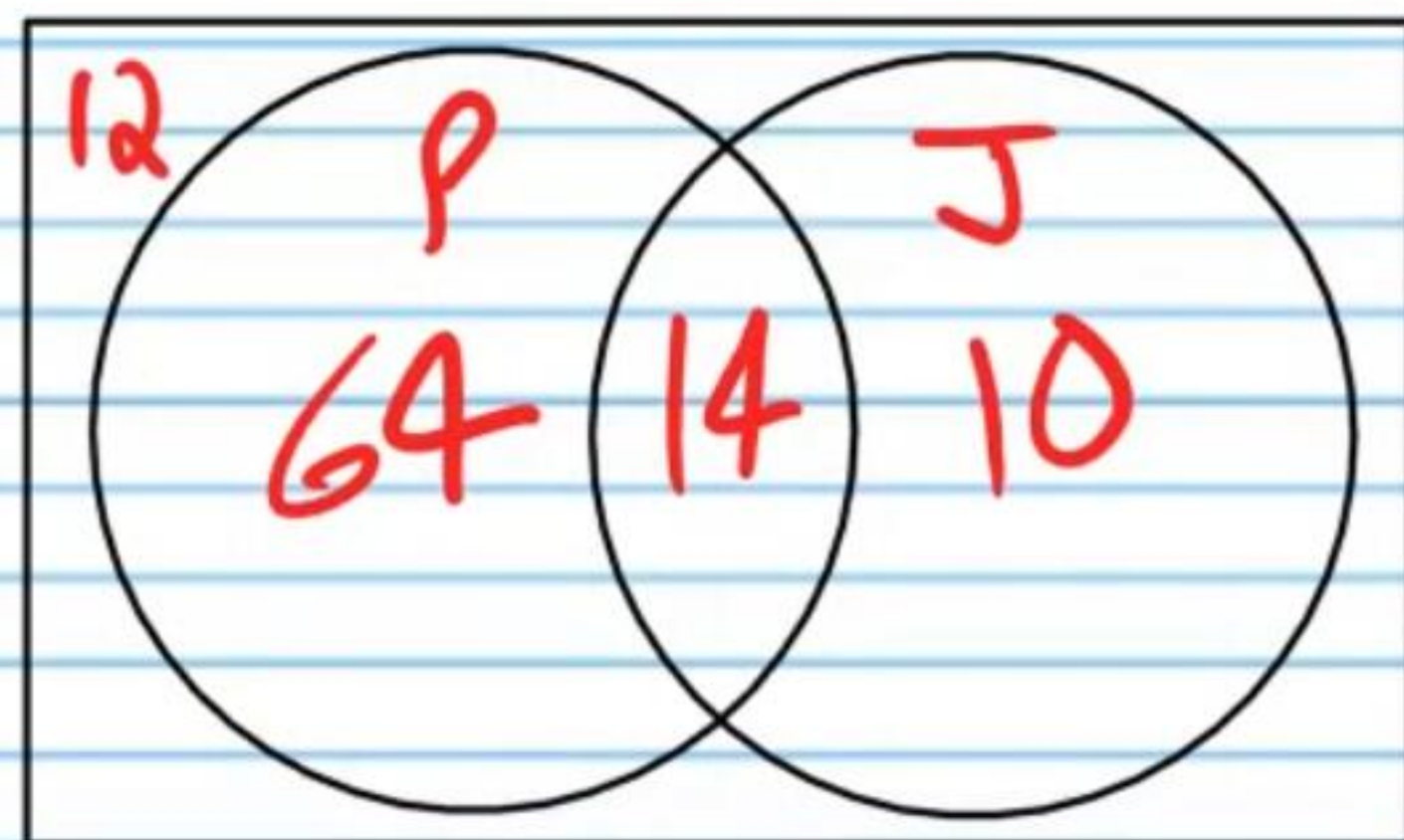
$$24 - 14 = \boxed{10}$$

iii) How many students like **either** peanut butter **or** jelly?

$$64 + 14 + 10 = \boxed{88}$$

iv) How many students do not like peanut butter or jelly?

$$100 - 88 = \boxed{12}$$

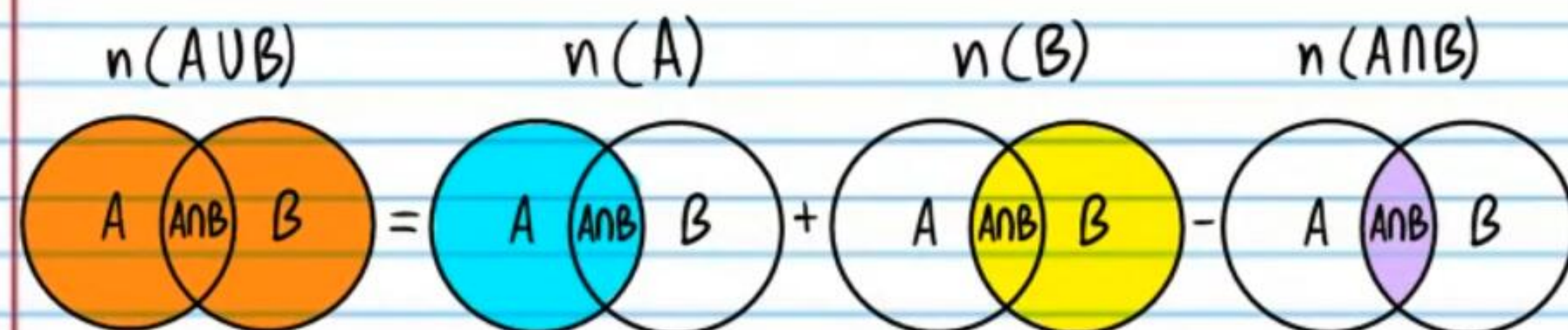


If "A" and "B" are sets, then $n(A)$ is the number of things in "A" and $n(B)$ is the number of things in "B". $n(A \cap B)$ is the number of things in **both** "A" **and** in "B", and $n(A \cup B)$ is the number of things in **either** "A" **or** "B".

Counting the Number of Things in **either** Set "A" **or** Set "B".

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Explanation



- (23) A school requires every student to choose at least one area of science to study. There are 350 students at the school. 200 of them chose **either** biology **or** physics. 110 of them chose biology and 136 chose physics.

i) How many students chose neither physics nor biology?

$$350 - 200 = \boxed{150}$$

ii) How many students chose both physics and biology?

$$n(B \cup P) = n(B) + n(P) - n(B \cap P)$$

$$200 = 110 + 136 - n(B \cap P)$$

$$200 = 246 - n(B \cap P)$$

$$-46 = -n(B \cap P)$$

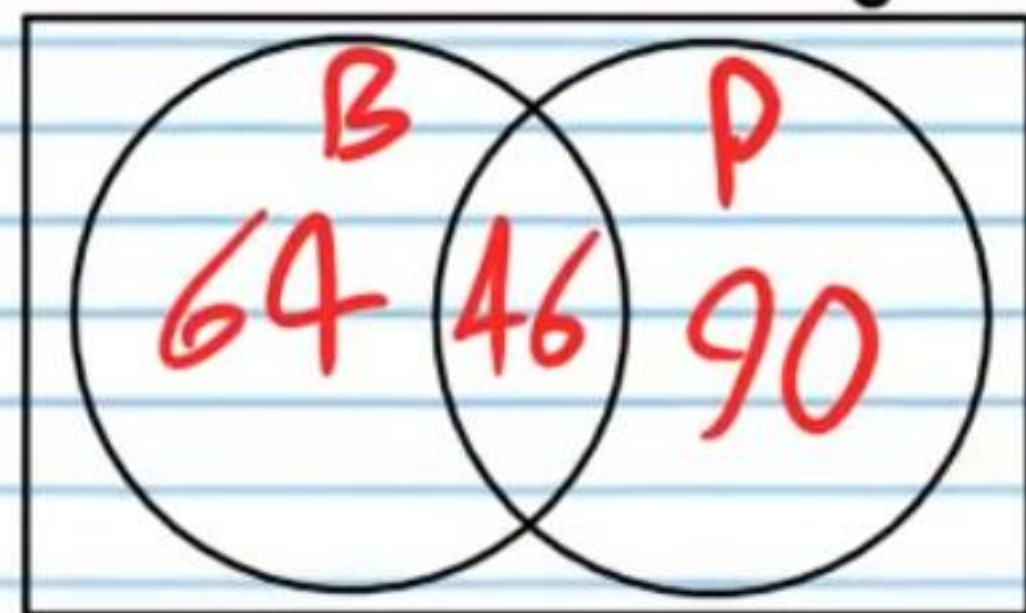
$$\boxed{46} = n(B \cap P)$$

iii) How many students chose biology but not physics?

$$110 - 46 = \boxed{64}$$

iv) How many students chose physics but not biology?

$$136 - 46 = \boxed{90}$$



- (24) A travel agency checks its records and finds that 24 of its customers have been to Spain and 40 have been to France. 49 have been to **either** Spain **or** France. The agency has 77 customers in total.

i) How many customers have been to neither Spain nor France?

$$77 - 49 = \boxed{28}$$

ii) How many customers have been to both Spain and France?

$$n(S \cup F) = n(S) + n(F) - n(S \cap F)$$

$$49 = 24 + 40 - n(S \cap F)$$

$$49 = 64 - n(S \cap F)$$

$$-15 = -n(S \cap F)$$

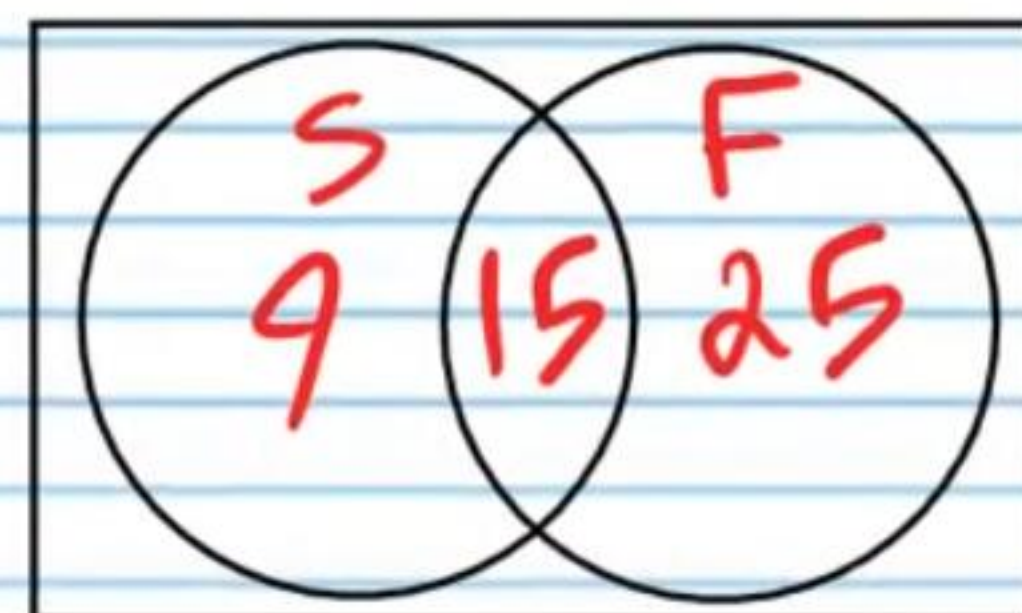
$$\boxed{15} = n(S \cap F)$$

iii) How many customers traveled to Spain but not France?

$$24 - 15 = \boxed{9}$$

iv) How many customers traveled to France but not Spain?

$$40 - 15 = \boxed{25}$$



- (25) Madeleine has 46 coins. Most of them are Spanish coins. Some of them are gold. 29 of them are **either** Spanish **or** gold. 21 are Spanish and 10 are gold.

i) How many coins are not gold or Spanish?

$$46 - 29 = \boxed{17}$$

ii) How many of Madeleine's coins are Spanish and also made of gold?

$$n(S \cup G) = n(S) + n(G) - n(S \cap G)$$

$$29 = 21 + 10 - n(S \cap G)$$

$$29 = 31 - n(S \cap G)$$

$$-2 = -n(S \cap G)$$

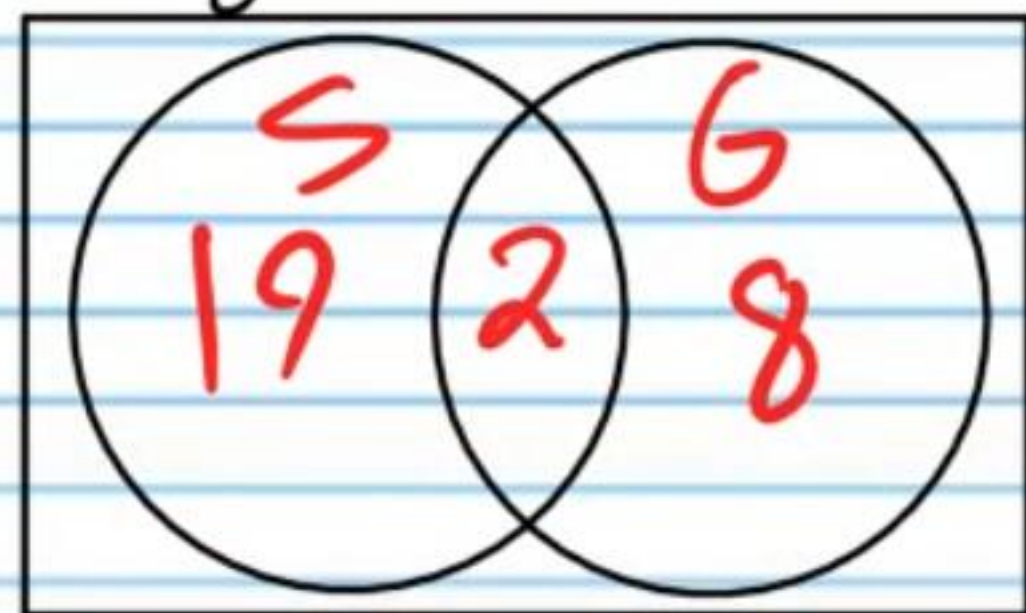
$$\boxed{2} = n(S \cap G)$$

iii) How many coins are Spanish but not made of gold?

$$21 - 2 = \boxed{19}$$

iv) How many coins are made of gold but are not Spanish?

$$10 - 2 = \boxed{8}$$



- (26) A restaurant serves 18 types of food. 9 types have wheat in the ingredients and 12 types have beef. 16 types have **either** wheat **or** beef.

i) How many types of food contain neither wheat nor beef?

$$18 - 16 = \boxed{2}$$

ii) How many types of food contain both wheat and beef?

$$n(W \cup B) = n(W) + n(B) - n(W \cap B)$$

$$16 = 9 + 12 - n(W \cap B)$$

$$16 = 21 - n(W \cap B)$$

$$-5 = -n(W \cap B)$$

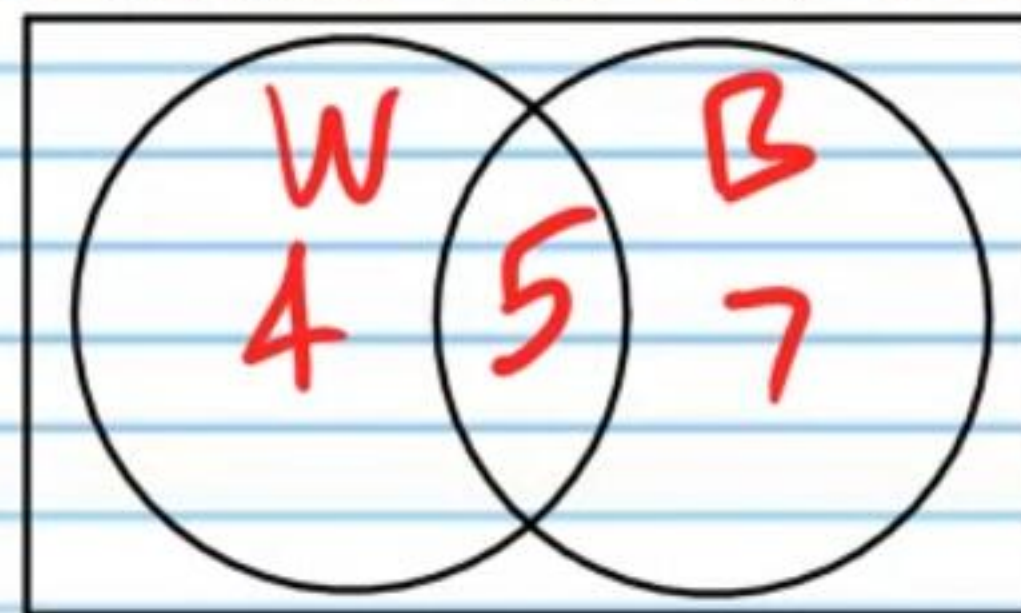
$$\boxed{5} = n(W \cap B)$$

iii) How many types of food contain wheat but not beef?

$$9 - 5 = \boxed{4}$$

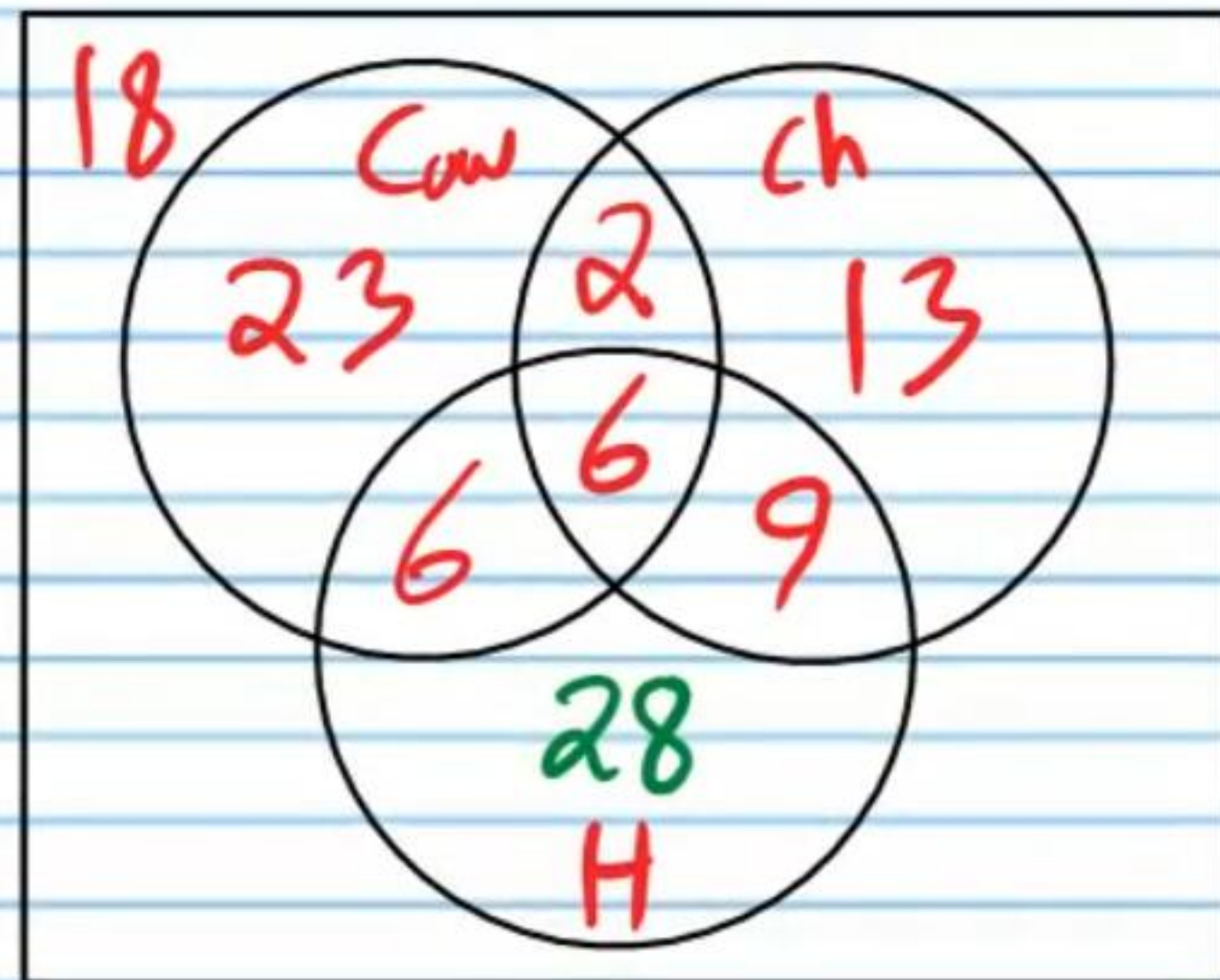
iv) How many types of food contain beef but not wheat?

$$12 - 5 = \boxed{7}$$



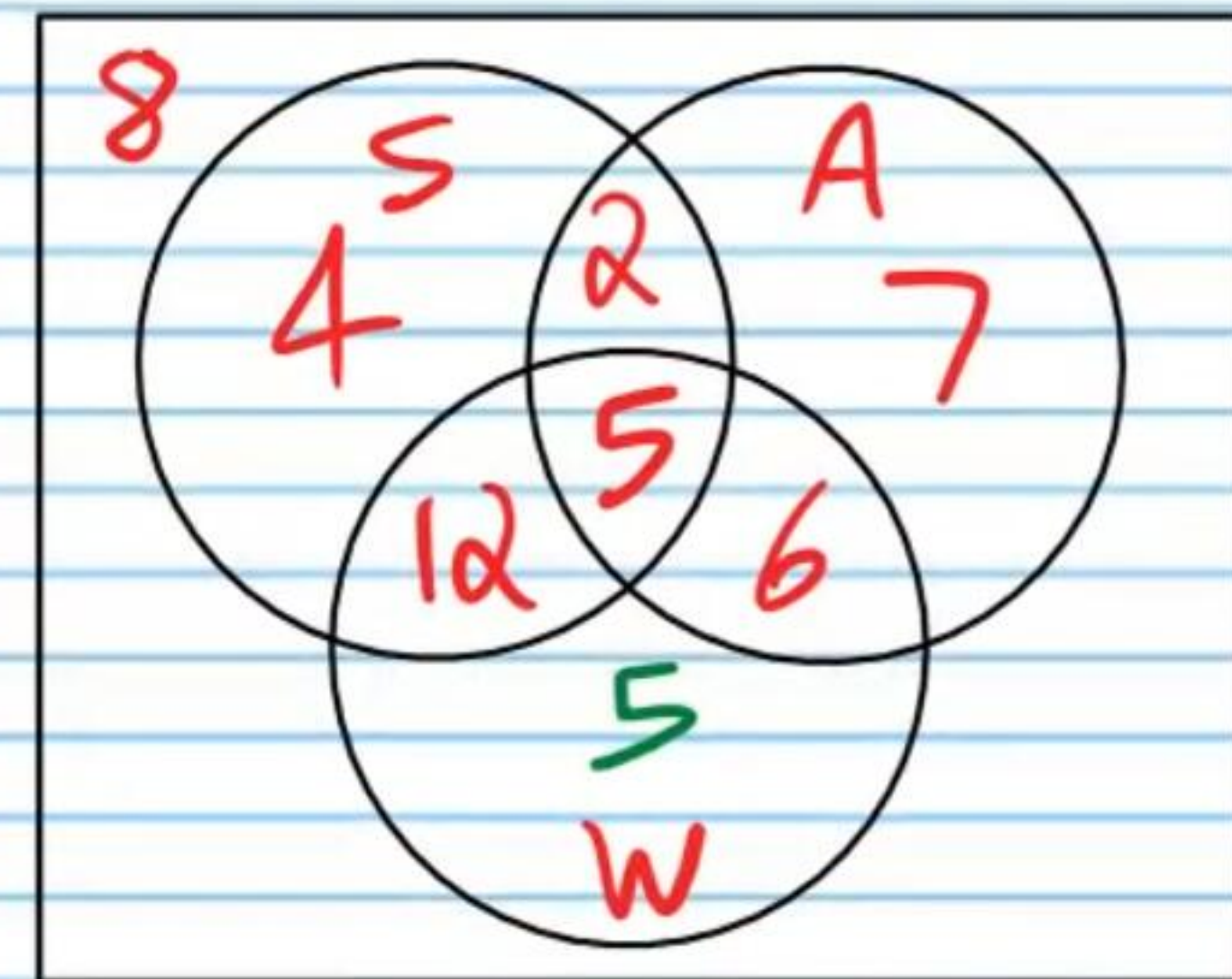
- (27) There are 105 farms in the county of Petersburg. 37 of them have cows, 30 of them have chickens, and 15 of them have horses and chickens. 12 of them have horses and cows. 8 of them have cows and chickens. 6 of them have all three. 18 farms do not have cows, chickens, or horses. How many farms have horses only?

$$105 - 18 - 23 - 13 - 6 - 6 - 2 - 9 = \boxed{28}$$



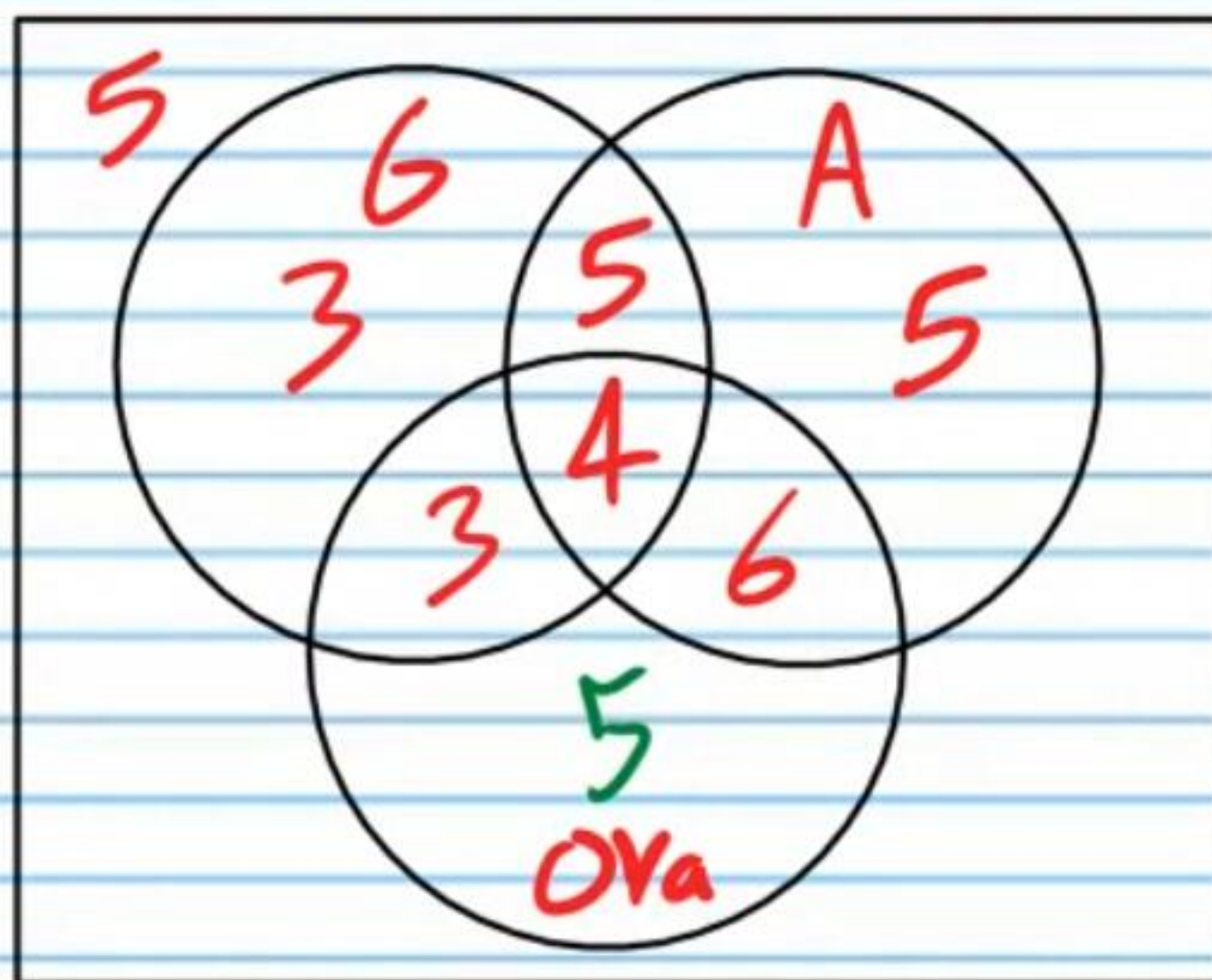
- (28) There are 49 homes in the town of Louisville. 23 of them have solar panels and 20 of them are air conditioned. 11 homes have a water well and air conditioning. 17 homes have a well and solar panels. 7 homes have solar panels and air conditioning, 5 homes have all three of these and 8 homes have none of them. How many homes have a well but no air conditioning and no solar panels?

$$49 - 8 - 4 - 2 - 5 - 12 - 7 - 6 = \boxed{5}$$



- (29) A business sends fruit baskets for gift-givers. They sent 36 baskets this month. 15 of the baskets contained grapes. 20 of the baskets contained apples. 10 of the baskets contained apples and oranges. 7 of the baskets contained oranges and grapes. 9 of the baskets contained grapes and apples. 4 baskets contain all three types of fruit and 5 baskets had none of these types of fruit. How many baskets had oranges but no apples or grapes?

$$36 - 5 - 3 - 5 - 3 - 4 - 5 - 6 = \boxed{5}$$



- (30) A pizza parlor made 85 pizzas this week. 54 of those pizzas had pepperonis on them. 39 of them had mushrooms. 21 of them had sausage and mushrooms. 25 of them had pepperonis and mushrooms. 20 of them had pepperonis and sausage. 13 pizzas had all three of these toppings. 16 pizzas had none of these toppings. How many pizzas had sausage but no pepperonis or mushrooms?

$$85 - 16 - 22 - 12 - 13 - 7 - 6 - 8 = \boxed{1}$$

